

AWS EC2 Task - 1

1. Launch one EC2 using Amazon Linux 2 image and add a script in user data to install Apache.

Step 1:

- Log in to the AWS Management Console.
- Navigate to EC2.
- Click Launch Instance.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

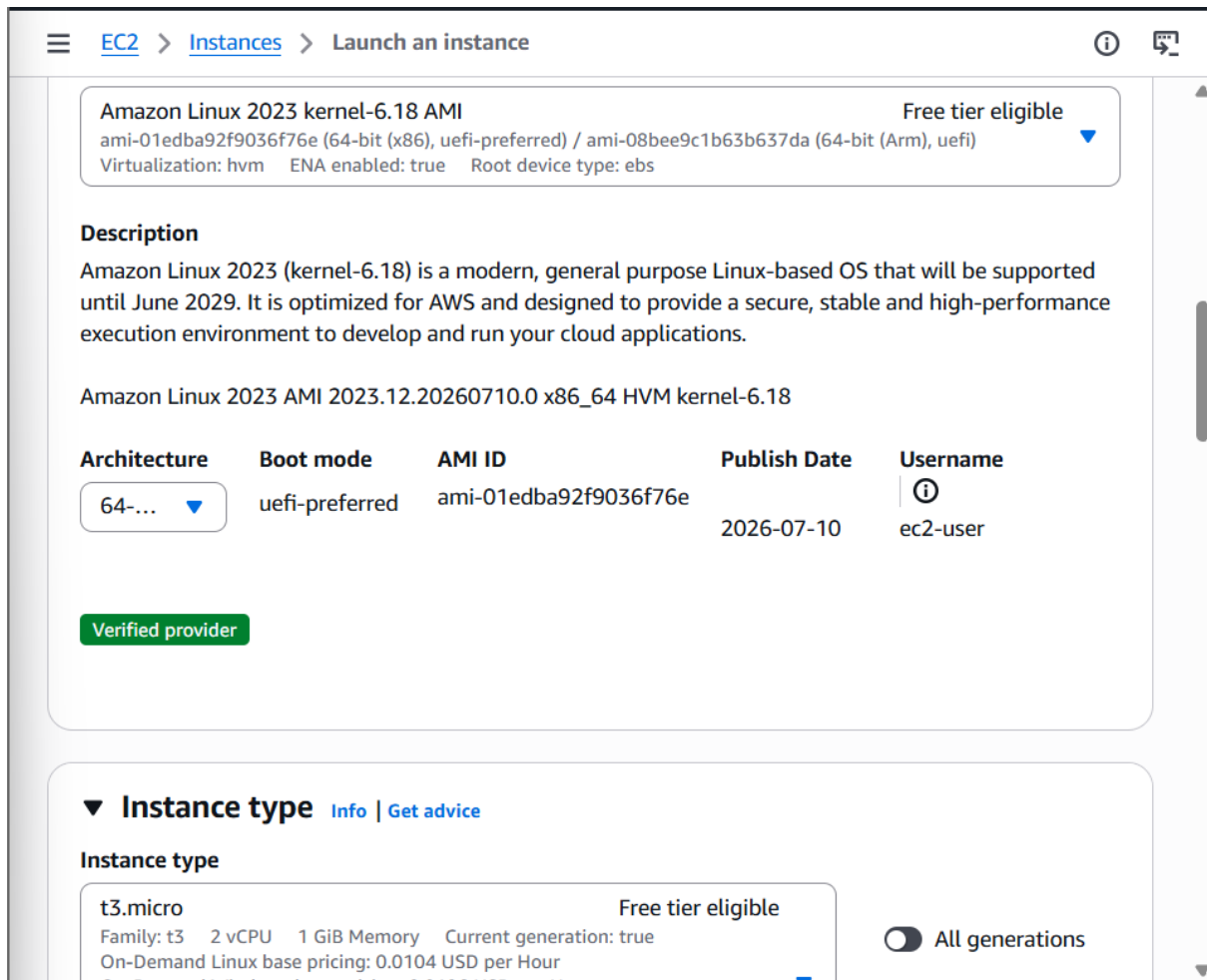
Red Hat

 [Browse more AMIs](#)

Step 2:

- Enter the **Instance Name**.
- Select **recent one**

- Choose the **t3.micro** (or any supported) instance type.
- Select an existing **Key Pair**.



Step 3:

- Configure the **Network Settings**.
- Enable **Auto-assign Public IP**.
- Select or create a **Security Group**.
- Allow **SSH (Port 22)** from **My IP**.
- Allow **HTTP (Port 80)** from **Anywhere (0.0.0.0/0)**.

EC2 > Instances > Launch an instance

Create key pair

Key pair name
Key pairs allow you to connect to your instance securely.

first_server

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ **RSA**
RSA encrypted private and public key pair

☐ **ED25519**
ED25519 encrypted private and public key pair

Private key file format

☒ **.pem**
For use with OpenSSH

☐ **.ppk**
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn more](#)

Cancel Create key pair

Step 4:

- Expand **Advanced Details**.
- Scroll down to the **User Data** section.

EC2 > Instances > Launch an instance

Allow tags in metadata

User data - optional

Upload a file with your user data or enter it in the field.

Choose file

```
#!/bin/bash
sudo yum -y install httpd
echo "hello installed using userdata" >> /var/www/html/index.html
sudo systemctl start httpd
```

☐ User data has already been base64 encoded

Summary

Number of instances: 1

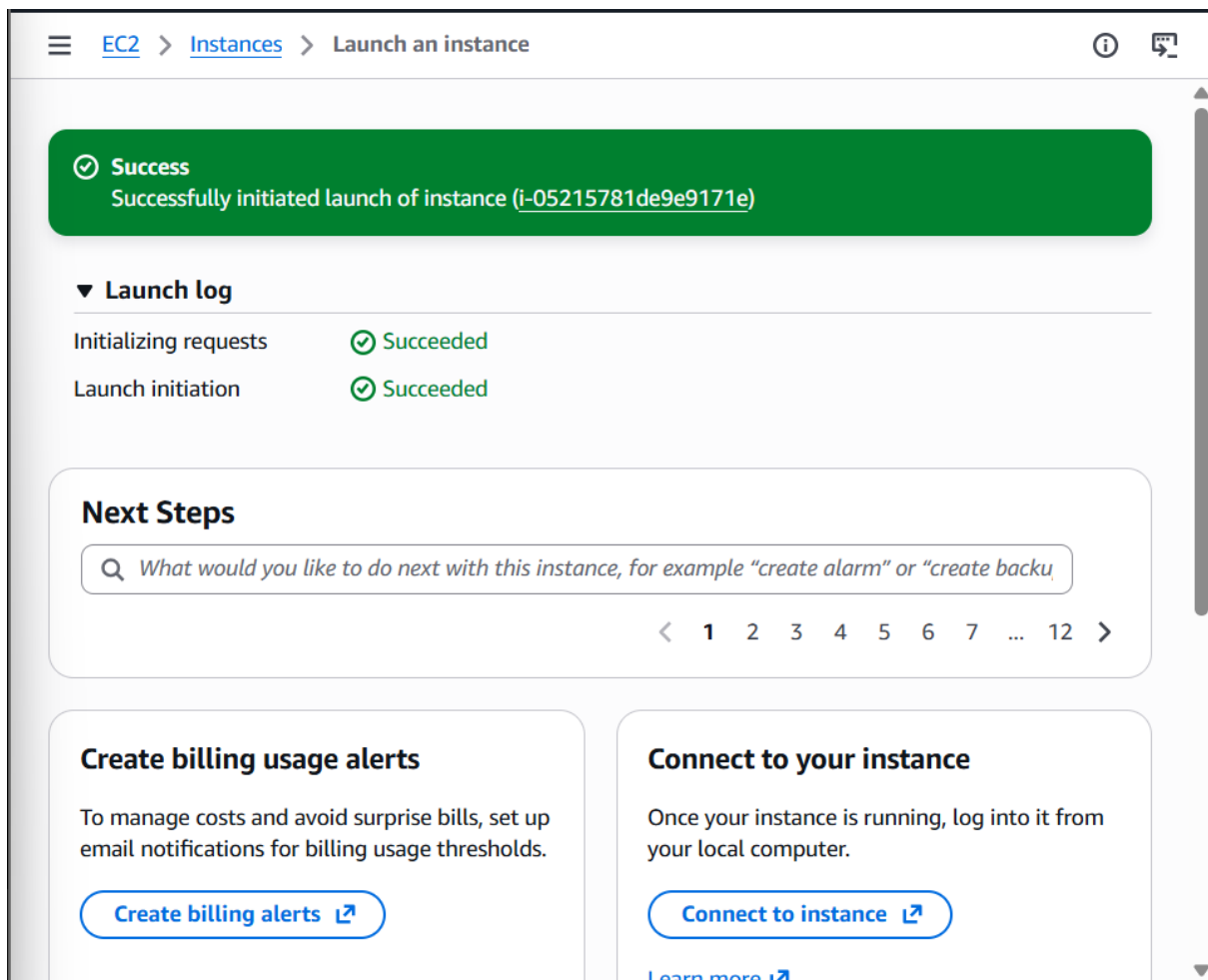
Software image (AMI)
Amazon Linux 2023 AMI 2023.12...read more
ami-01edba92f9036f76e

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Launch instance Preview code



Step 5:

- Click **Launch Instance**.
- Wait until the instance state changes to **Running**.
- Copy the **Public IPv4 Address**.
- Open a web browser and enter `http://<Public-IP>`.
- Verify that the Apache web page is displayed successfully.

Amazon Linux 2023

<https://aws.amazon.com/linux/amazon-linux-2023>

```
cat: index.html: No such file or directory
[ec2-user@ip-172-31-27-211 ~]$ cd /var/www/html
[ec2-user@ip-172-31-27-211 html]$ cat index.html
hello installed using userdata
[ec2-user@ip-172-31-27-211 html]$
```

2. Launch one EC2 using Ubuntu image and add a script in user data to install Nginx.

- Log in to the **AWS Management Console**.
- Navigate to **EC2**.
- Click **Launch Instance**.

- Enter the **Instance Name**.
- Select the **Ubuntu Server** AMI.

- Choose the **t3.micro**
- Select an existing **Key Pair**.

▼ **Application and OS Images (Amazon Machine Image)** [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Q Search our full catalog including 1000s of application and OS images

Recents Quick Start

Amazon Linux
aws

macOS
Mac

Ubuntu
Canonical Ubuntu

Windows
Microsoft

Red Hat
Red Hat

SUSE Linux
SUSE

Debian
debian

>


[Browse more AMIs](#)
 Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 26.04 LTS (HVM), SSD Volume Type
 ami-0b6d9d3d33ba97d99 (64-bit (x86)) / ami-0bc7f2dbdcc6b5303 (64-bit (Arm))
 Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

Description

Step 3:

- Configure the **Network Settings**.
- Enable **Auto-assign Public IP**.
- Select or create a **Security Group**.
- Allow **SSH (Port 22)** from **My IP**.
- Allow **HTTP (Port 80)** from **Anywhere (0.0.0.0/0)**.

Step 4:

- Expand **Advanced Details**.
- Scroll down to the **User Data** section.

User data - *optional* | [Info](#)

Upload a file with your user data or enter it in the field.

 [Choose file](#)

```
#!/bin/bash

apt update -y

apt install -y nginx

echo "Hello installed using userdata - Nginx Server" > /var/www/html/index.html

systemctl start nginx

systemctl enable nginx
```

☐ User data has already been base64 encoded

Step 5:

- Click **Launch Instance**.
- Wait until the instance state changes to **Running**.
- Copy the **Public IPv4 Address**.
- Open a web browser and enter `http://<Public-IP>`

```

brajk@RajKumari MINGW64 ~/Downloads
$ ssh -i first_server.pem ubuntu@54.208.204.43
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Jul 14 12:39:48 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   33.9% of 6.61GB Processes:      119
Memory usage: 28%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 172.31.20.1

Expanded Security Maintenance for Applications is not enabled.

68 updates can be applied immediately.
50 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-172-31-20-1:~$ systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-07-14 12:31:18 UTC; 9min ago
   Invocation: cb67c7341dbf4b23b88c7aead27ff182
     Docs: man:nginx(8)
  Main PID: 1537 (nginx)
    Tasks: 3 (limit: 627)
   Memory: 3M (peak: 7.1M)
      CPU: 80ms
   CGroup: /system.slice/nginx.service
           └─1537 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─1539 "nginx: worker process"
               └─1540 "nginx: worker process"

Jul 14 12:31:18 ip-172-31-20-1 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jul 14 12:31:18 ip-172-31-20-1 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-20-1:~$

```

← → ↻ ⚠ Not secure 54.208.204.43

Hello installed using userdata - Nginx Server

3. Launch one Windows server and install Tomcat on Windows.

Objective

Launch a Windows EC2 instance on AWS and install Apache Tomcat to access the Tomcat web server from a browser.

Step 1:

- Log in to the AWS Management Console.
- Navigate to EC2.
- Click Launch Instance.

Step 2:

- Enter the **Instance Name**.
- Select the **Microsoft Windows Server AMI**.
- Choose the **t3.micro**
- Select an existing **Key Pair**.

EC2

Instances

Launch an instance

Name

Window-Tomcat

Add additional tags

▼ Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Q

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

Canonical Ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

>

Q

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Microsoft Windows Server 2025 Base

ami-09639480113b0df96 (64-bit (x86))

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

Step 4:

- Click **Launch Instance**.
- Wait until the instance state changes to **Running**.
- Select the instance and click **Connect** → **RDP Client**.
- Retrieve and decrypt the Windows Administrator password using your **.pem** key.
- Connect to the instance using **Remote Desktop Connection (mstsc)**

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#instances?v=3&instanceState=running&sort=ipv6-addresses

aws [Search] [Alt+S] United States (N. Virginia) Rajkumari (571833581790) Rajkumari

EC2 > Instances

Instances (1/1) Info Last updated less than a minute ago [Connect] [Instance state] [Actions] [Launch instances]

Saved filter sets Choose filter set Find Instance by attribute or tag (case-sensitive)

Instance state = running [Clear filters]

☒	Name	Instance ID	Instance state	Instance type	
☒	Window-Tomcat	i-0b289e32a21943c80	Running	t3.micro	

Change security groups
Modify IAM role
Get Windows password

Instance diagnostics
Instance settings
Networking
Security
Image and templates
Storage
Monitor and troubleshoot

i-0b289e32a21943c80 (Window-Tomcat)

Details Status and alarms Monitoring Security Networking Storage Tags

▼ Instance summary Info

Windows

Get Windows password

Connect to your Windows instance using Remote Desktop with this information.

Instance ID
i-0b289e32a21943c80 (Window-Tomcat)

Private IP address
172.31.28.45

Username
Administrator

Password
=KClxjxlb.Q7%k0!fQ-angA.dxy)i17

Password change recommended
We recommend that you change your default password. Note: If a default password is changed, it cannot be retrieved using this tool. It is important that you change your password to one that you will remember.

Cancel OK

SSM Session Manager

RDP client

EC2 serial console

Instance ID

 i-0b289e32a21943c80 (Window-Tomcat)

Connection Type



Connect using RDP client

Download a file to use with your RDP client and retrieve your password.



Connect using Fleet Manager

To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be installed and running on the instance. For more information, see [Working with SSM Agent](#) 

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

 [Download remote desktop file](#)

When prompted, connect to your instance using the following username and password:

Public DNS

 ec2-44-223-32-201.compute-1.amazonaws.com

Username [Info](#)



Administrator



Password [Get password](#)

▼  RajKumari - Personal

>  Attachments

 window_key.pem

14-07-2026 18:26

PEM File

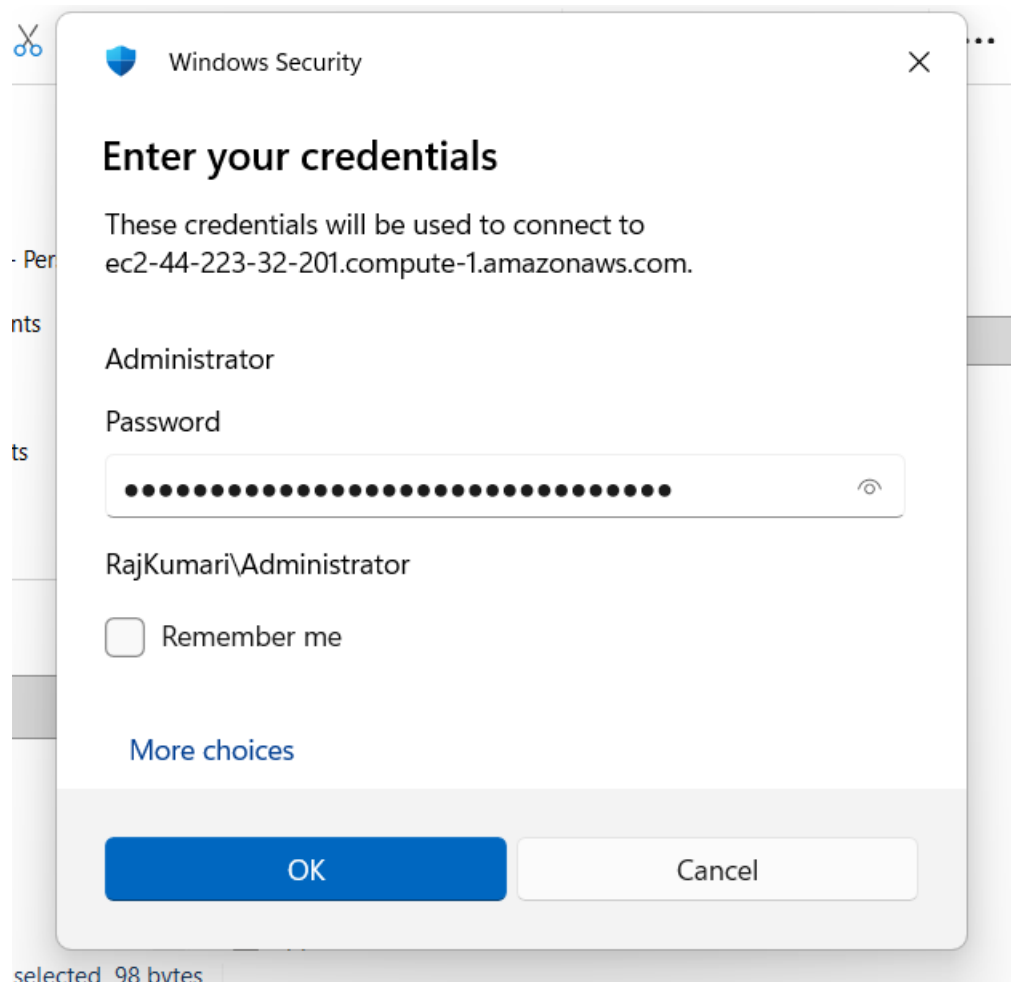
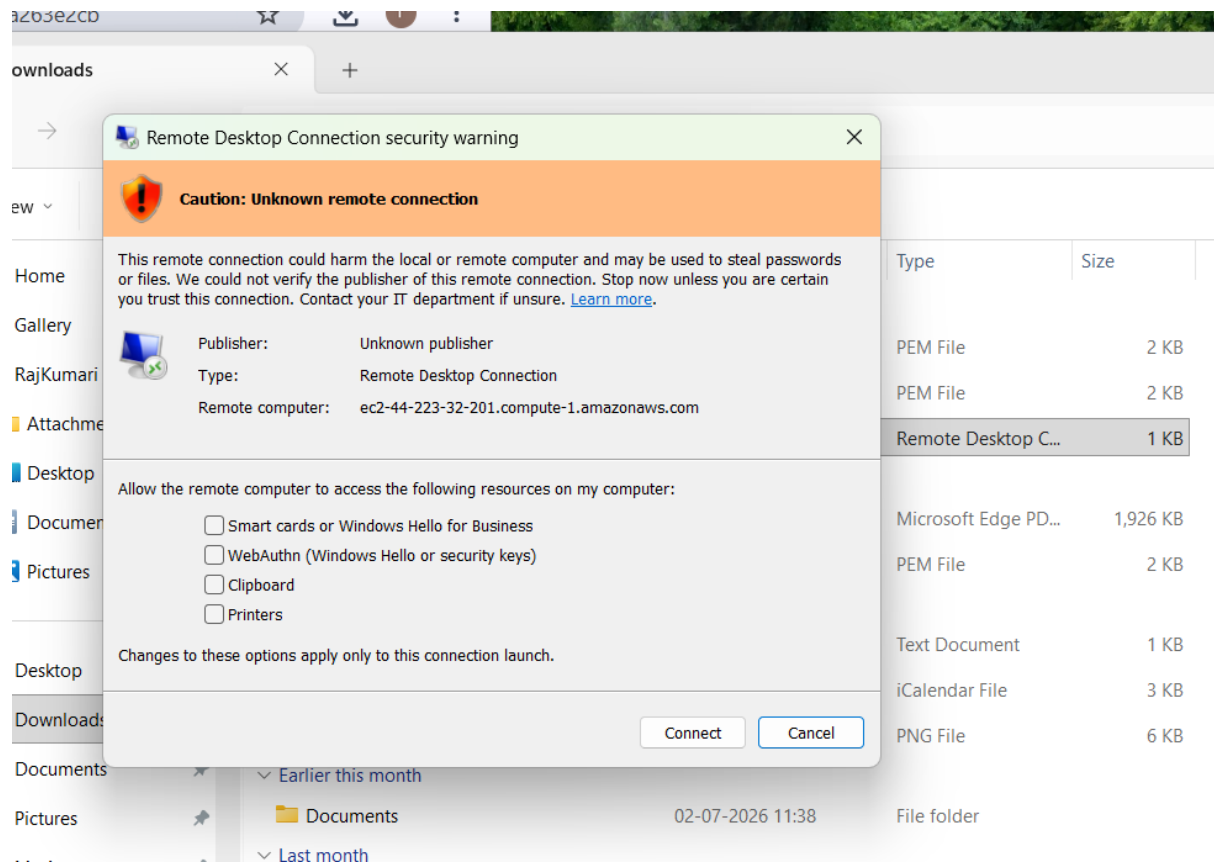
2 KB

 Window-Tomcat

14-07-2026 18:42

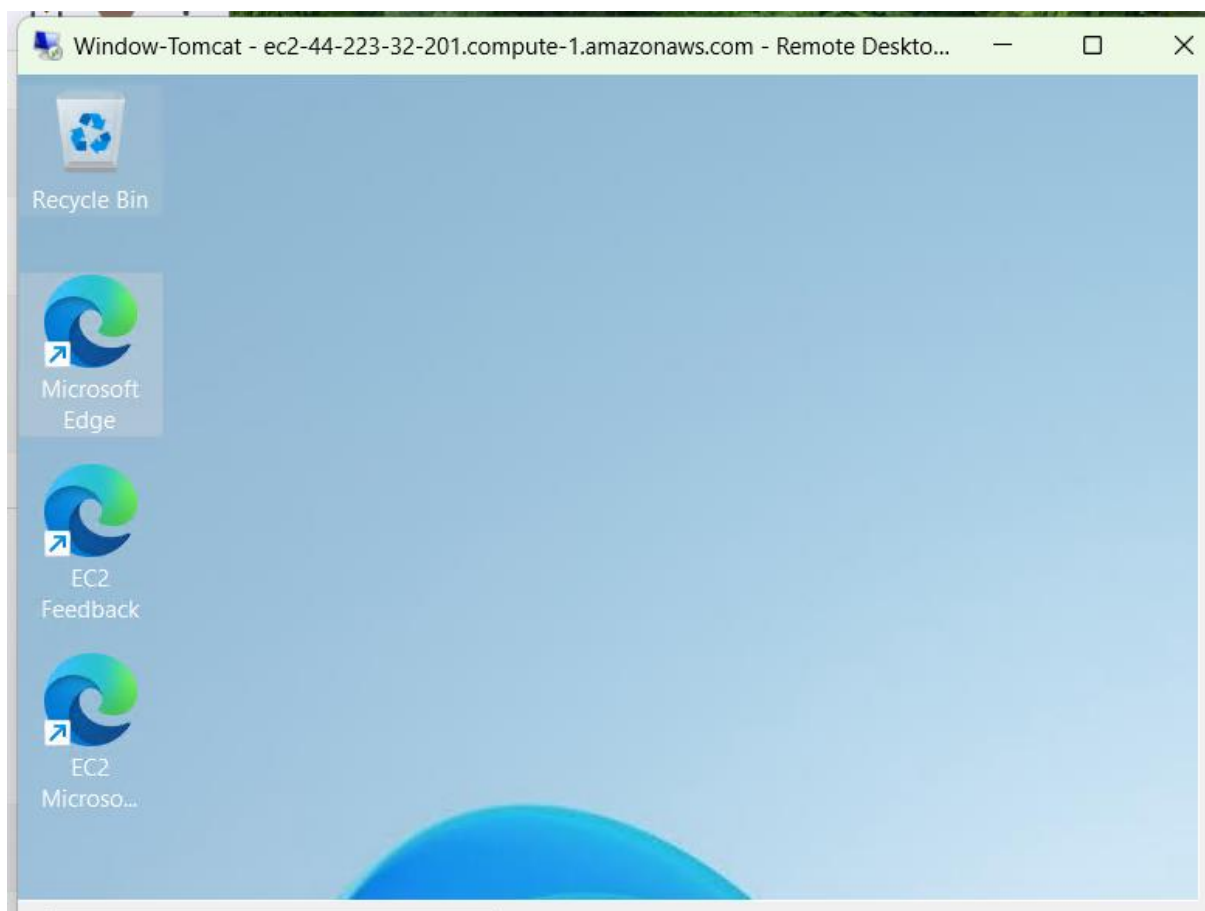
Remote Desktop C...

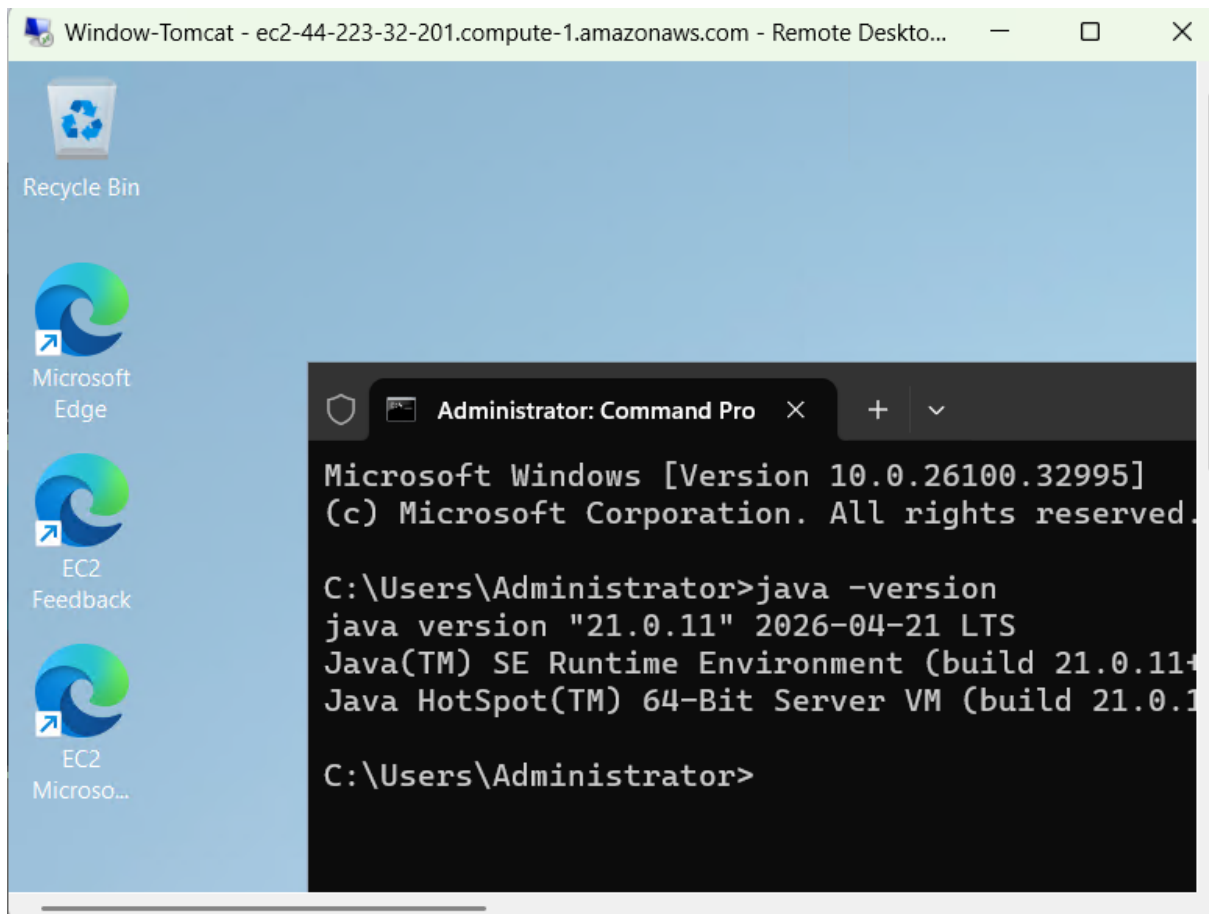
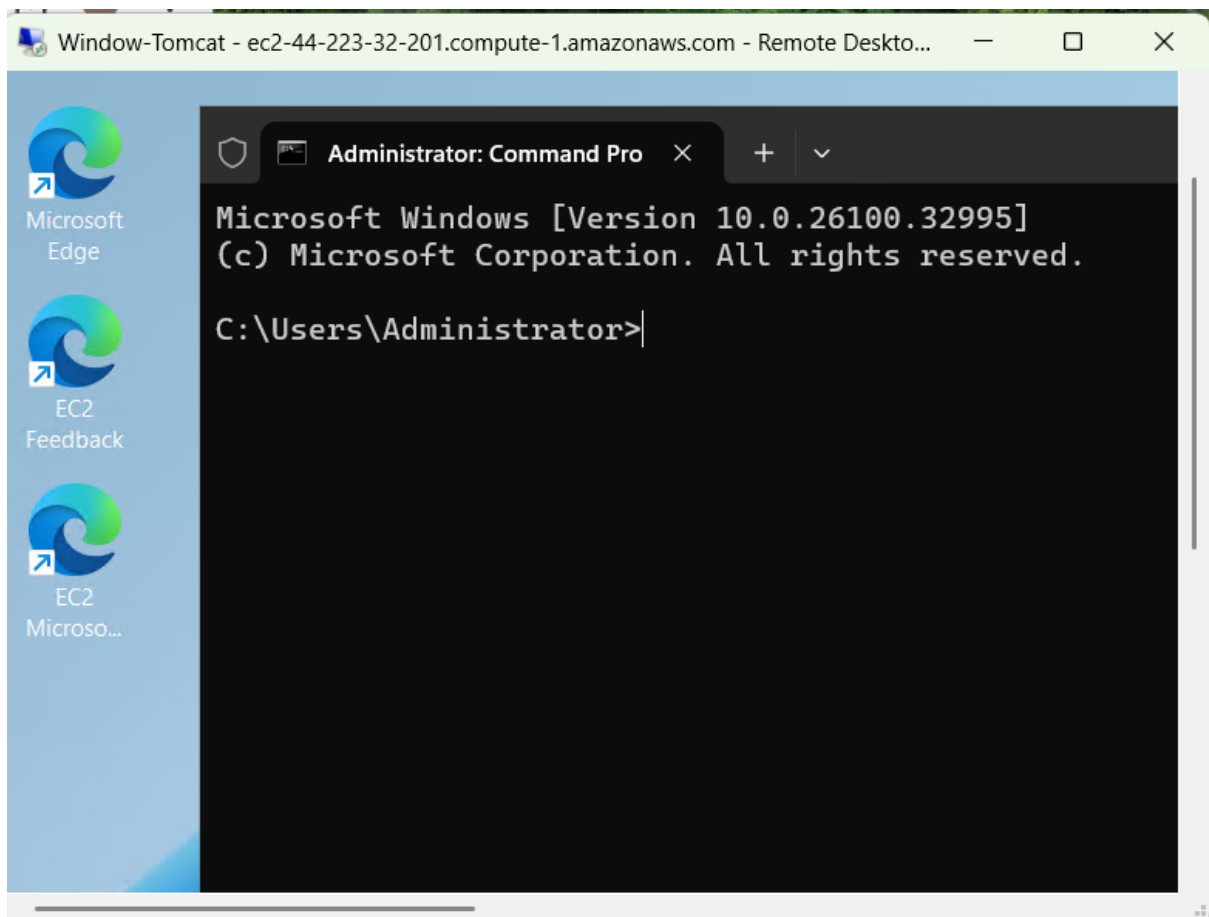
1 KB

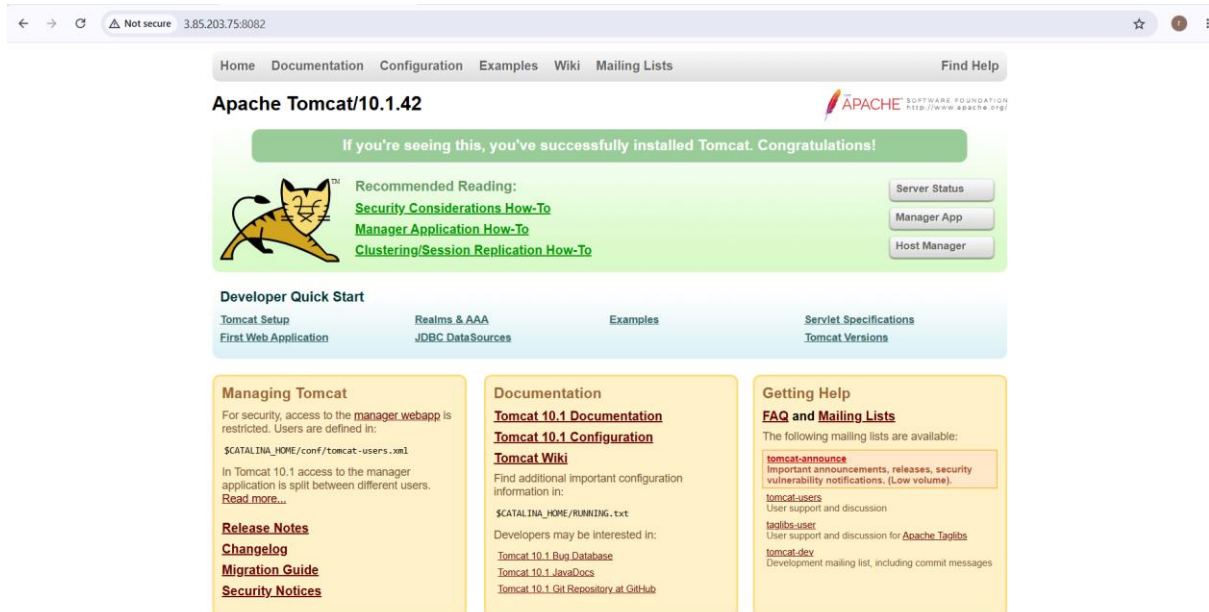


Step 5:

- Download and install **Java JDK**.
- Set the **JAVA_HOME** environment variable.
- Download the **Apache Tomcat** ZIP file from the official website.
- Extract Tomcat (for example, to C:\Apache\apache-tomcat-10.x.x).
- Start Tomcat by running startup.bat from the **bin** folder.
- Open a web browser and access <http://<Public-IP>:8080> to verify that the Tomcat home page is displayed successfully.





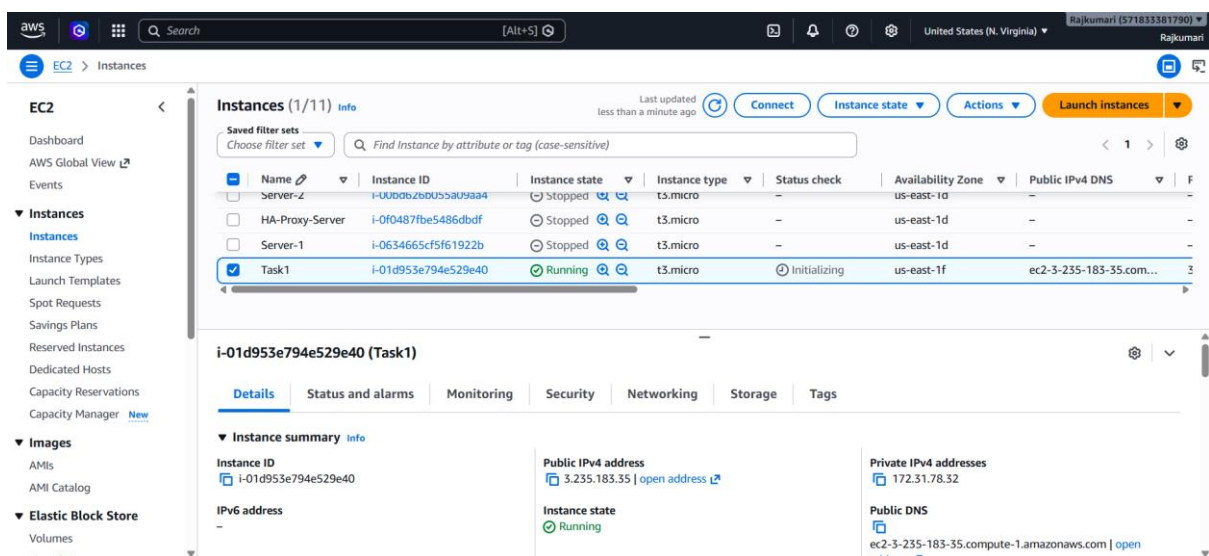


4. Take a snapshot of the instance created in Task 1.

Objective: Create a snapshot (backup) of the EC2 instance's EBS volume.

Step 1:

- Log in to the AWS Management Console.
- Navigate to EC2.
- Select the EC2 instance created in Task 1.



Step 2:

- Go to the **Storage** tab.
- Select the attached **EBS Volume**.
- Click on the **Volume ID** to open the EBS volume details.

The screenshot shows the AWS Management Console interface for EBS volumes. At the top, a green notification bar states: "Successfully created snapshot snap-06c01fd5da02a3e3d from volume vol-0598ee564351c440d. If you need your snapshot to be immediately available consider using Fast Snapshot Restore." To the right of this bar is a button labeled "Manage fast snapshot restore". Below the notification, the "Volumes (1)" section is active, showing a table with one volume. The volume ID "vol-0598ee564351c440d" is highlighted in the table. Below the table, there is a "Fault tolerance for all volumes in this Region" section. At the bottom, a "Snapshot summary" section is visible, indicating that 0 out of 11 volumes were backed up in the last 24 hours.

Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume ID	Created
	vol-0598ee564351c440d	gp3	8 GiB	3000	125	snap-0ace8a0...	-	2026/

Step 3:

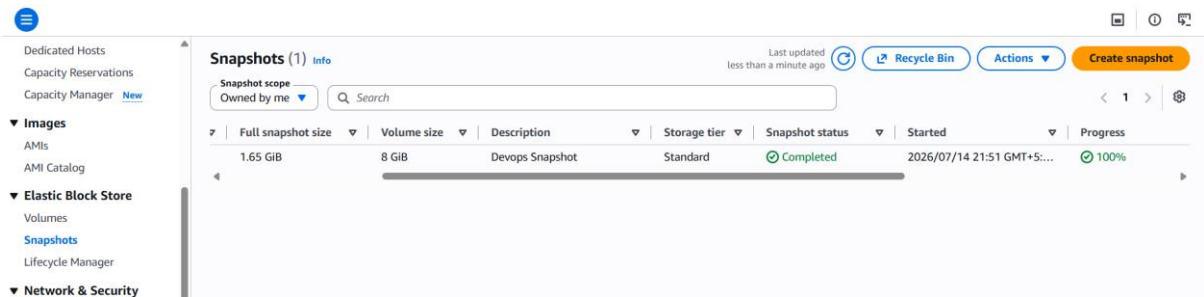
- Select the EBS volume.
- Click **Actions** → **Create Snapshot**.

Step 4:

- Enter a **Snapshot Name** and **Description** (optional).
- Click **Create Snapshot**.

Step 5:

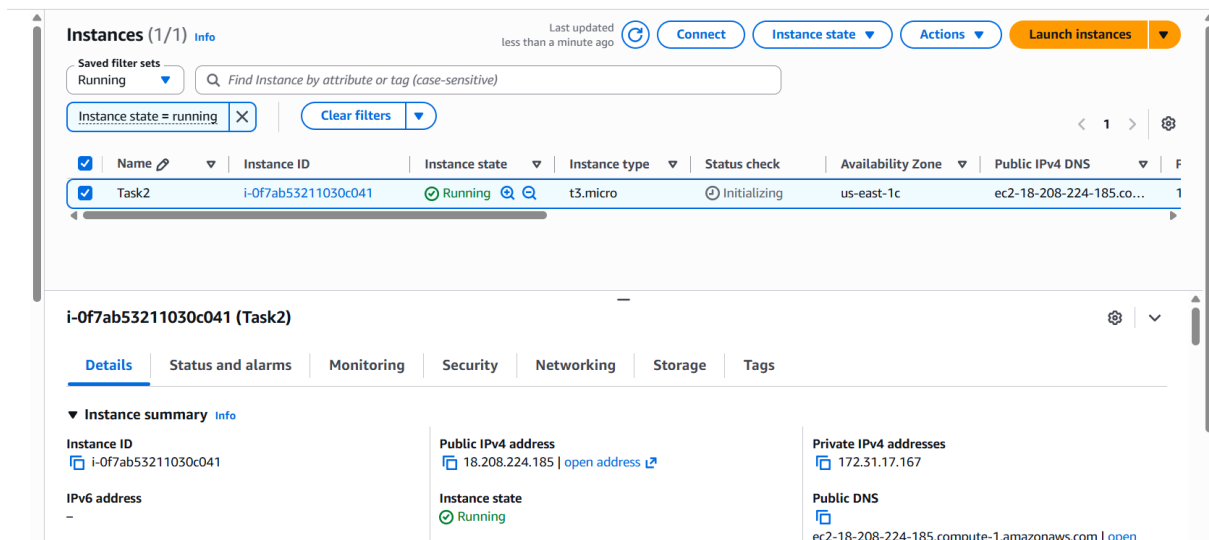
- Navigate to **Elastic Block Store** → **Snapshots**.
- Verify that the snapshot status changes to **Completed**, confirming that the snapshot was created successfully



5.Assign passwordless authentication for the EC2 created in Task 2.

Step 1:

- Connect to the Amazon Linux EC2 instance using the existing **.pem** key.
- Open a terminal on the instance.



Step 2:

- Generate a new SSH key pair by running **ssh-keygen**.
- Press **Enter** to accept the default file location.
- Press **Enter** twice to leave the passphrase empty.

```

brajk@RajKumari MINGW64 ~/Downloads
$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/c/Users/brajk/.ssh/id_rsa):
/c/Users/brajk/.ssh/id_rsa already exists.
Overwrite (y/n)? y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/brajk/.ssh/id_rsa
Your public key has been saved in /c/Users/brajk/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:y4BSBZdwJ1x6YDNcBTlae5r91ghmlBfDrR5Vi+++96w brajk@RajKumari
The key's randomart image is:
+---[RSA 3072]---+
| .*B@+o. . . |
| +*oB+ o. . |
| o.o...=. . |
| ....+.+. . |
| .*.oS. . |
| .o =o.. . |
| o ooo . |
| + .. .. |
| . ooEoo |
+-----[SHA256]-----+
brajk@RajKumari MINGW64 ~/Downloads
$ |

```

Step 3:

- Display the public key using `cat ~/.ssh/id_rsa.pub`.
- Append the public key to the `~/.ssh/authorized_keys` file
- Set the required permissions:
 - `chmod 700 ~/.ssh`
 - `chmod 600 ~/.ssh/authorized_keys`

```
[ec2-user@ip-172-31-17-167 ~]$ ls -la ~/.ssh
total 4
drwx-----, 2 ec2-user ec2-user 29 Jul 14 16:35 .
drwx-----, 3 ec2-user ec2-user 74 Jul 14 16:35 ..
-rw-----, 1 ec2-user ec2-user 394 Jul 14 16:35 authorized_keys
[ec2-user@ip-172-31-17-167 ~]$ ssh-keygen
-bash: ssh-keygen: command not found
[ec2-user@ip-172-31-17-167 ~]$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/ec2-user/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ec2-user/.ssh/id_rsa
Your public key has been saved in /home/ec2-user/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:IXZiUoHnvXBuo8hRmdcFVg5Z0h5E7YlN+mzBcLN832Y ec2-user@ip-172-31-17-167.ec2.internal
The key's randomart image is:
+---[RSA 3072]-----+
|
|  =B=
|  ..+=
|  = 0.0% +
|  . * = 0+ 0 .
|  . B . S 0 0 ..
|  + = + E
|  . + 0 . 0
|  . 0 +0.
|  o .oo.
+---[SHA256]-----+
[ec2-user@ip-172-31-17-167 ~]$ AC
[ec2-user@ip-172-31-17-167 ~]$ cat /home/ec2-user/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGCg4KXtoGWIksHmJsunnmjuERPzoZRwajuUvXyg+M6yS2G6IsaD0gk1a1hd5KbHQTC
XKn9SrEG4UbIouBd2kzPNmrAvm2PKPvtWJZF12UthWpZ2VI4nEvZbrhxAQUk5rptHVT45kPEskEbTW6VeQyPngxHLcemBFbhAuT71zb
FSAYnFjYgNh+pVldGtZ0eB3WiyCwBZxtKCPBcW54HA5YGjdSLMKA1FeBtV3d9Y2ZEYrGa5+dSY/dI1KBDoCd7DWZQ41PmnF3RmXrgWK
sDTJNfnk7voUvi7QNPChjdp7e48b1I8Uske51+pGBcC1q9NUYfBpj1k8BTBVZXL7jQtAw3M2bVQECF634JhoKcP+1KjXjFoLqut+1n
rUmIrFjowrTOKkPwd4/RAeFPMlVFarlNRIFZ81JiTeGcAxAsiQonFIvMVUkuH9XnrggTE77QGEw6Ej+gf2NMP1ng98wM0S23j3obXZv
7CyeuHczC9GnHe/zdyQ6TGsbGScw4I7dtfvM= ec2-user@ip-172-31-17-167.ec2.internal
[ec2-user@ip-172-31-17-167 ~]$
```

Step 4:

- Display the private key using `cat ~/.ssh/id_rsa`.
- Copy the entire private key.
- Save it on your local machine as `id_rsa` inside the `~/.ssh` directory (for Windows Git Bash: `C:\Users\<username>\.ssh\id_rsa`)

Step 5:

- Ensure the private key has the correct permissions on your local machine (if required).
- Connect to the Amazon Linux EC2 instance **without using the .pem file**:

`$ ssh ec2-user@<Public-IP>`

- Verify that you can log in successfully without specifying the `.pem` key, confirming that passwordless authentication has been configured successfully.

EC2 > Instances > Launch an instance

Instance bandwidth configuration | Info

Select ▼

Purchasing option | Info

☐ None

☐ Capacity Blocks
Launch instances for your active capacity blocks

☐ Interruptible Capacity Reservations
Launch instances for your interruptible Capacity Reservations

☒ Spot instances
Request Spot Instances at the Spot price, capped at the On-Demand price

[Customize Spot instance options](#)

Capacity reservation | Info

Select ▼

Tenancy | Info

Select ▼

RAM disk ID | Info

Select ▼

Step 5: Configure Network Settings

- Choose your VPC and subnet (or leave the defaults).
- Enable Auto-assign Public IP.
- Allow SSH (Port 22) from your IP address.

Step 6: Configure Storage

Step 7: Launch the Instance

Click Launch instance.

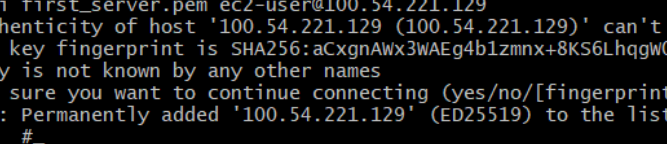
Step 8: Verify It's a Spot Instance

After the instance is running:

1. Go to EC2 → Instances.
2. Select your instance.

Lifecycle: Spot

```
brajk@RajKumari MINGW64 ~/Downloads
$ ssh -i first_server.pem ec2-user@100.54.221.129
The authenticity of host '100.54.221.129 (100.54.221.129)' can't be established.
ED25519 key fingerprint is SHA256:aCxgnAwX3wAEg4blZmnx+8KS6LhqgwOaodXL5hrH75c.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '100.54.221.129' (ED25519) to the list of known hosts.
```



```
#_
#####
#####|
\##/
V'-'>
~-.
~/m/'
```

Amazon Linux 2023
<https://aws.amazon.com/linux/amazon-linux-2023>

```
[ec2-user@ip-172-31-28-87 ~]$ |
```

7. Enable termination policy on the EC2 created in Task 2.

Using the AWS Console

1. Sign in to the AWS Management Console.
2. Go to EC2 → Instances.
3. Select the EC2 instance created in Task 2.
4. Click Actions → Instance settings → Change termination protection.
5. Select: Enable
6. Click Save.

EC2 > Instances

EC2

Dashboard

AWS Global View

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

▼ Images

AMIs

AMI Catalog

▼ Elastic Block Store

Volumes

1 minute ago

Actions

Launch instances

Instance settings

Protection and lifecycle management

Attach to Auto Scaling Group

Change stop protection

Change termination protection

Change shutdown behavior

Change auto-recovery behavior

Change instance migration on reboot

Protect with Elastic Disaster Recovery

Instance configuration

Change instance type

Change CPU options

Modify instance placement

Clear filters

1

Public IPv4 ...

1-129.co... 100.54.221.129

7-218.co... 54.227.177.218

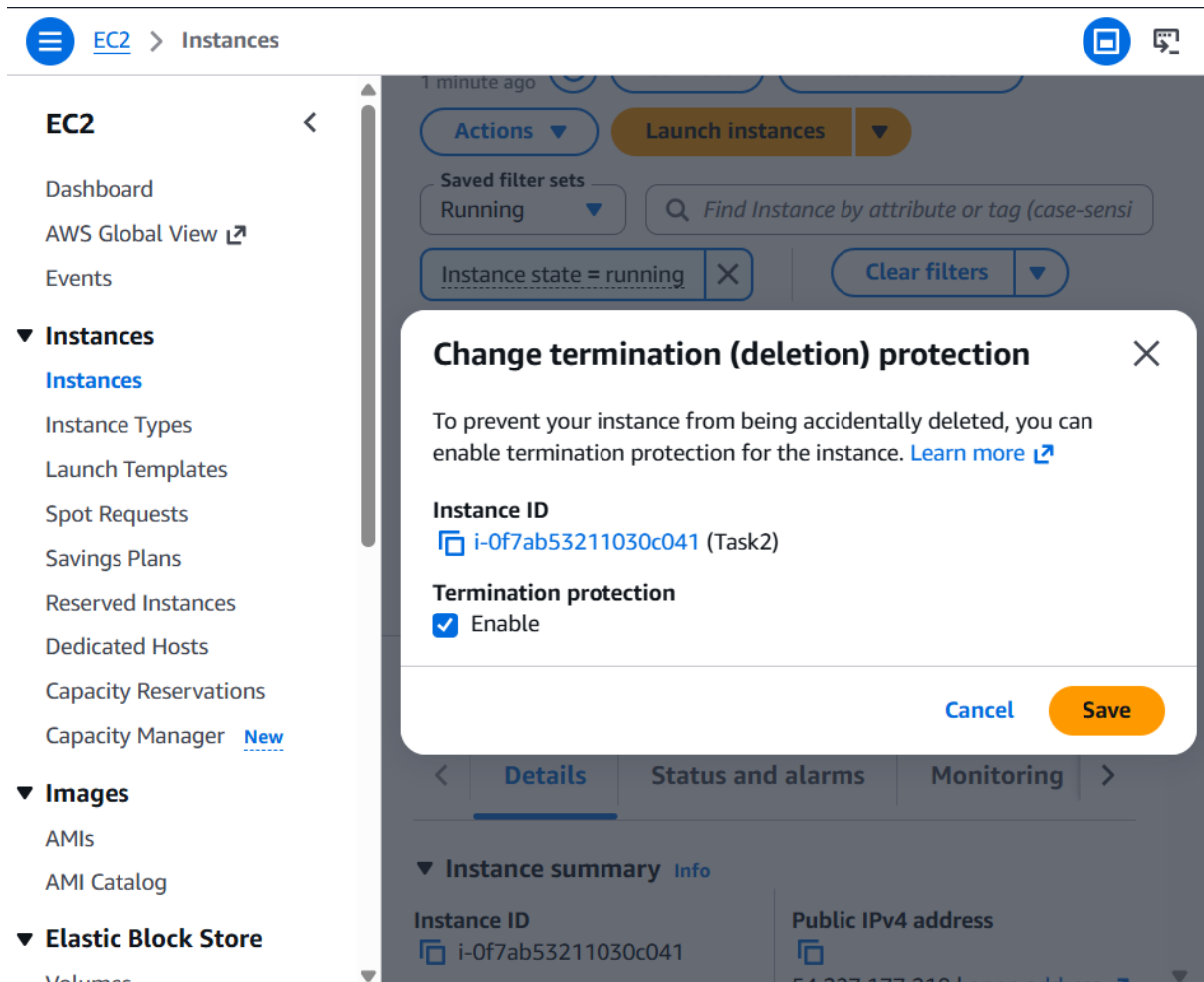
1

arms

Monitoring

public IPv4 address

54.227.177.218 | [Launch address](#)



Verify

1. Select the instance.
2. Open the Details tab.
3. Look for: Termination protection: Enabled

i-0f7ab53211030c041 (Task2)



▼ Instance details [Info](#)

AMI ID ami-01edba92f9036f76e	Monitoring disabled	Platform details Linux/UNIX
AMI name al2023-ami-2023.12.20260710.0-kernel-6.18-x86_64	Allowed image -	Termination protection Enabled
Stop protection Disabled	Launch time Tue Jul 14 2026 23:07:52 GMT+0530 (India Standard Time) (4 minutes)	AMI location amazon/al2023-ami-2023.12.20260710.0-kernel-6.18-x86_64
Instance reboot migration Default (On)	Instance auto-recovery Default	Lifecycle normal

8. Launch one EC2 using AWS CLI.

Step 1: Download AWS CLI

1. Open your web browser.
2. Go to the official AWS CLI download page:

<https://aws.amazon.com/cli/>

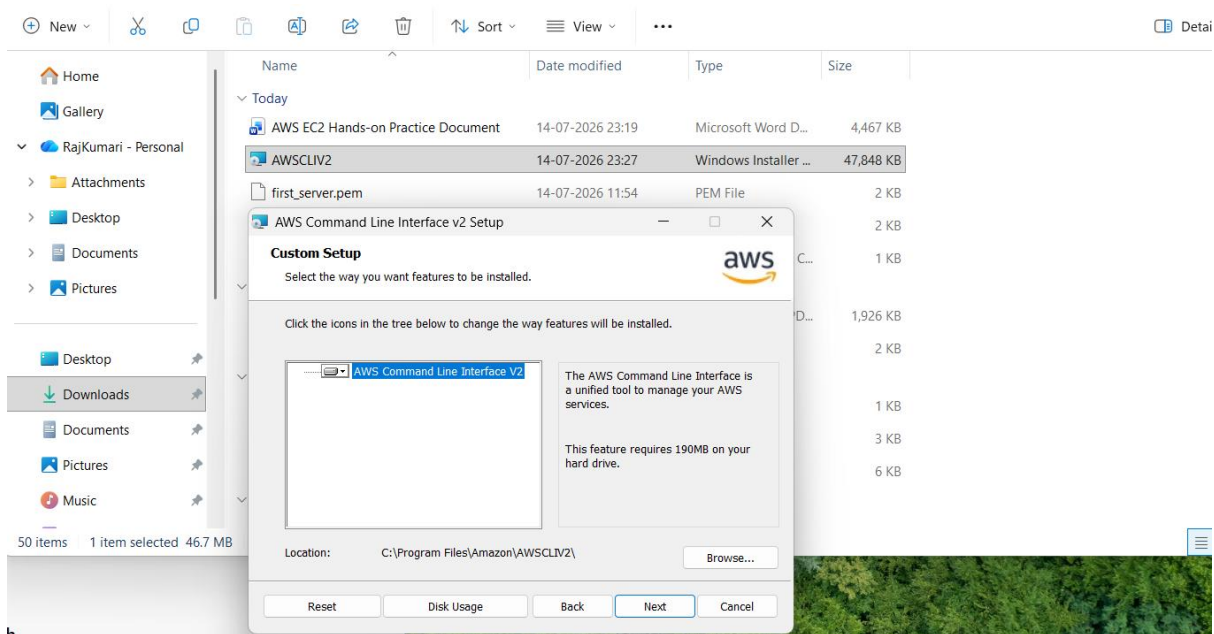
3. Click Download AWS CLI for Windows (64-bit).

Or download the MSI installer directly:

<https://awscli.amazonaws.com/AWSCLIV2.msi>

Step 2: Install AWS CLI

1. Double-click AWSCLIV2.msi.
2. Click Next.
3. Click Next again.
4. Click Install.
5. Wait for the installation to complete.
6. Click Finish.



Step 3: Verify Installation

Open Command Prompt, PowerShell, or Git Bash.

Run:

aws --version

```
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\brajk>aws --version
aws-cli/2.35.22 Python/3.14.6 Windows/11 exe/AMD64

C:\Users\brajk>
```

Step 4: Create an IAM User (Skip if you already have Access Keys)

If you already have an Access Key ID and Secret Access Key, skip to Step 5.

Otherwise:

1. Log in to the AWS Management Console.

2. Go to IAM.
3. Click Users.
4. Click Create User.
5. Enter a username (for example, awscli-user).
6. Attach a policy such as AmazonEC2FullAccess (for learning purposes).
7. Create the user.
8. Open the user.
9. Go to Security credentials.
10. Under Access keys, click Create access key.
11. Choose Command Line Interface (CLI).
12. Create the access key.
13. Download or copy the:
 - Access Key ID
 - Secret Access Key

Important: Save the Secret Access Key securely. You won't be able to view it again later.

IAM > IAM users > Create user ⓘ

✔ User created successfully

You can view and download the user's password and email instructions for signing in to the AWS Management Console.

[View user](#)

Step 1
Specify user details

Step 2
Set permissions

Step 3
Review and create

Step 4
Retrieve password

Retrieve password

You can view and download the user's password below or email users instructions for signing in to the AWS Management Console. This is the only time you can view and download this password.

Console sign-in details

[Email sign-in instructions ↗](#)

Console sign-in URL

<https://571833381790.signin.aws.amazon.com/console>

User name

[rajkumari](#)

Console password

[Show](#)

[Cancel](#) [Download .csv file](#) [Return to users list](#)

IAM > IAM users > [rajkumari](#) > Create access key ⓘ

► Step 1 of 3

Access key best practices & alternatives [Info](#)

Avoid using long-term credentials like access keys to improve your security. Consider the following use cases and alternatives.

Use case

☒ **Command Line Interface (CLI)**
You plan to use this access key to enable the AWS CLI to access your AWS account.

☐ **Local code**
You plan to use this access key to enable application code in a local development environment to access your AWS account.

☐ **Application running on an AWS compute service**

Step 5: Configure AWS CLI

Run:

aws configure

You'll be prompted for:

AWS Access Key ID:

AWS Secret Access Key:

Default region name:

Default output format:

Example:

AWS Access Key ID: AKIA*****

AWS Secret Access Key: *****

Default region name: us-east-1

Default output format: json

Purpose: Stores your AWS credentials locally.

Step 6: Verify the Configuration

Run:

aws configure list

Expected output:

Name	Value	Type

access_key	*****ABCD	config-file
secret_key	*****WXYZ	config-file
region	us-east-1	config-file

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\brajk>aws --version
aws-cli/2.35.22 Python/3.14.6 Windows/11 exe/AMD64

C:\Users\brajk>aws configure

Tip: You can deliver temporary credentials to the AWS CLI using your AWS Console session by running the command 'aws login'.

AWS Access Key ID [None]: AKIAYKI7M60PPZD327GJ
AWS Secret Access Key [None]: TRwB0Jvw7YiRnQ2eaLXIR/P4zouRBDGHTNWgIpcj
Default region name [None]: us-east-1
Default output format [None]: json

C:\Users\brajk>aws configure list
NAME      : VALUE                                : TYPE                : LOCATION
profile    : <not set>                               : None                : None
access_key : *****27GJ                             : shared-credentials-file :
secret_key : *****lpcj                             : shared-credentials-file :
region     : us-east-1                               : config-file         : ~/.aws/config

C:\Users\brajk>
```

Step 7: Verify Your AWS Account

Run:

aws sts get-caller-identity

```
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\brajk>where more
C:\Windows\System32\more.com

C:\Users\brajk>aws sts get-caller-identity
{
  "UserId": "AIDAYKI7M60PFEHIVCGC3",
  "Account": "571833381790",
  "Arn": "arn:aws:iam::571833381790:user/rajkumari"
}

C:\Users\brajk>
```

Step 8: List Available Key Pairs

Run:

aws ec2 describe-key-pairs --output table

```
C:\Users\brajk>aws ec2 describe-key-pairs --output table
```

DescribeKeyPairs	
KeyPairs	
CreateTime	2026-06-30T16:54:15.074000+00:00
KeyFingerprint	b6:3e:7d:26:75:cc:da:ca:db:d2:a9:f8:e8:20:27:dd:03:3d:cb:36
KeyName	ha-proxy
KeyPairId	key-0889420a8fbcedef1
KeyType	rsa
KeyPairs	
CreateTime	2026-07-14T06:24:44.543000+00:00
KeyFingerprint	41:26:b5:b3:52:8f:d6:16:64:85:3d:91:cc:af:17:61:af:60:72:b3
KeyName	first_server
KeyPairId	key-05524497d15050015
KeyType	rsa
KeyPairs	

```
-- More --
```

Step 9: List Security Groups

aws ec2 describe-security-groups --query

"SecurityGroups[*].[GroupId,GroupName]" --output table

```
C:\Users\brajk>aws ec2 describe-security-groups --query "SecurityGroups[*].[GroupId,GroupName]" --output table
```

DescribeSecurityGroups	
sg-0f09cf8a21a42ab47	launch-wizard-2
sg-06c27de9e09f09710	launch-wizard-3
sg-0d415660bb247f3ae	default
sg-01dace53c5fcef08e	launch-wizard-5
sg-021a7056b25c41fcf	launch-wizard-4
sg-033f1e4522502edbd	launch-wizard-1

```
C:\Users\brajk>
```

Step 10: Get the Latest Amazon Linux 2023 AMI

aws ssm get-parameters --names /aws/service/ami-amazon-linux-latest/al2023-ami-kernel-default-x86_64 --query "Parameters[0].Value" --output text

```
C:\Users\brajk>aws sts get-caller-identity
{
  "UserId": "AIDAYKI7M6OPFEHIVCGC3",
  "Account": "571833381790",
  "Arn": "arn:aws:iam::571833381790:user/rajkumari"
}

C:\Users\brajk>aws configure set cli_pager ""
C:\Users\brajk>aws sts get-caller-identity
{
  "UserId": "AIDAYKI7M6OPFEHIVCGC3",
  "Account": "571833381790",
  "Arn": "arn:aws:iam::571833381790:user/rajkumari"
}

C:\Users\brajk>aws ssm get-parameters --names /aws/service/ami-amazon-linux-latest/al2023-ami-kernel-d
efault-x86_64 --query "Parameters[0].Value" --output text
ami-0fd6240f599091088

C:\Users\brajk>
```

Step 3: Verify your key pair

Run: `aws ec2 describe-key-pairs --output table`

```
C:\Users\brajk>aws ec2 describe-key-pairs --output table
```

DescribeKeyPairs	
KeyPairs	
CreateTime	2026-06-30T16:54:15.074000+00:00
KeyFingerprint	b6:3e:7d:26:75:cc:da:ca:db:d2:a9:f8:e8:20:27:dd:03:3d:cb:36
KeyName	ha-proxy
KeyPairId	key-0889420a8fbcedf1
KeyType	rsa
KeyPairs	
CreateTime	2026-07-14T06:24:44.543000+00:00
KeyFingerprint	41:26:b5:b3:52:8f:d6:16:64:85:3d:91:cc:af:17:61:af:60:72:b3
KeyName	first_server
KeyPairId	key-05524497d15050015
KeyType	rsa
KeyPairs	
CreateTime	2026-07-14T12:56:31.904000+00:00
KeyFingerprint	ee:60:d0:e7:57:e2:ae:98:6e:15:56:a0:8a:8e:87:4f:c5:91:86:ab
KeyName	window_key
KeyPairId	key-012aa930c59c7c554
KeyType	rsa
KeyPairs	
CreateTime	2026-07-13T09:23:23.772000+00:00
KeyFingerprint	03:c4:68:30:0e:ab:3f:b8:ae:79:bb:42:33:05:0c:1b:14:16:9d:c2
KeyName	devops-key
KeyPairId	key-0ae7fcb37957326b4
KeyType	rsa
KeyPairs	
CreateTime	2026-06-26T18:24:45.769000+00:00
KeyFingerprint	32:e5:8a:01:ba:4c:6a:b7:7d:de:a5:13:a9:c9:0d:9b:ee:51:70:aa
KeyName	linux-testing
KeyPairId	key-0ea9f2c8968aae77b
KeyType	rsa
KeyPairs	
CreateTime	2026-06-30T16:52:11.101000+00:00
KeyFingerprint	2a:83:a8:75:03:9e:2d:24:6f:c1:11:8d:87:c5:36:e3:79:65:ee:39
KeyName	server-1
KeyPairId	key-02ad6a526955576d9
KeyType	rsa
KeyPairs	

Step 5: Launch the EC2 instance

Replace the AMI ID with the one you got in Step 2.

```
aws ec2 run-instances --image-id ami-0fd6240f599091088 --instance-  
type t3.micro --key-name first_server --security-group-ids sg-  
033f1e4522502edbd --count 1
```

```

    "StateReason": {
      "Code": "pending",
      "Message": "pending"
    },
    "VirtualizationType": "hvm",
    "CpuOptions": {
      "CoreCount": 1,
      "ThreadsPerCore": 2
    },
    "CapacityReservationSpecification": {
      "CapacityReservationPreference": "open"
    },
    "MetadataOptions": {
      "State": "pending",
      "HttpTokens": "required",
      "HttpPutResponseHopLimit": 2,
      "HttpEndpoint": "enabled",
      "HttpProtocolIpv6": "disabled",
      "InstanceMetadataTags": "disabled"
    },
    "EnclaveOptions": {
      "Enabled": false
    },
    "BootMode": "uefi-preferred",
    "PrivateDnsNameOptions": {
      "HostnameType": "ip-name",
      "EnableResourceNameDnsARecord": false,
      "EnableResourceNameDnsAAAARecord": false
    },
    "MaintenanceOptions": {
      "AutoRecovery": "default",
      "RebootMigration": "default"
    },
    "CurrentInstanceBootMode": "uefi",
    "Operator": {
      "Managed": false
    },
    "InstanceId": "i-0abc1d9f928e46279",
    "ImageId": "ami-0fd6240f599091088",
    "State": {
      "Code": 0,
      "Name": "pending"
    },
    "PrivateDnsName": "ip-172-31-12-141.ec2.internal",
    "PublicDnsName": "",
    "StateTransitionReason": "",
    "KeyName": "first_server",
    "AmiLaunchIndex": 0,
    "ProductCodes": [],
    "InstanceType": "t3.micro",
    "LaunchTime": "2026-07-14T18:58:19+00:00",
    "Placement": {
      "AvailabilityZoneId": "use1-az1",
      "GroupName": "",
      "Tenancy": "default",

```

Step 6: Verify the instance

Run: `aws ec2 describe-instances --query`

`"Reservations[*].Instances[*].[InstanceId,State.Name,PublicIpAddress]`
`" --output table`

```
C:\Users\brajk>aws ec2 describe-instances --query "Reservations[*].Instances[*].[InstanceId,State.Name,PublicIpAddress]" --output table
```

DescribeInstances		
i-0d3b2c8f9e663ad19	stopped	None
i-0bbc07b95a03b6824	stopped	None
i-08f366c947a19e340	stopped	None
i-0c10ec1b11c09e64b	stopped	None
i-0fdde7e929f9d9fb5	stopped	None
i-05215781de9e9171e	stopped	None
i-0b289e32a21943c80	stopped	None
i-0f7ab53211030c041	stopped	None
i-0abc1d9f928e46279	running	44.199.252.139
i-00bd626b055a09aa4	stopped	None
i-0f0487fbe5486dbdf	stopped	None
i-0634665cf5f61922b	stopped	None
i-01d953e794e529e40	stopped	None

```
C:\Users\brajk>
```

Connect to the instance

Once the instance is running and has a public IP:

```
ssh -i first_server.pem ec2-user@
```

Get Only the Public IP

If you know your instance ID: `aws ec2 describe-instances --instance-ids i-0abc1d9f928e46279 --query`

`"Reservations[0].Instances[0].PublicIpAddress" --output text`

```
C:\Users\brajk>aws ec2 describe-instances --instance-ids i-0abc1d9f928e46279
--query "Reservations[0].Instances[0].PublicIpAddress" --output text
44.199.252.139
```

```
C:\Users\brajk>
```

Step 7: Connect to the instance

Once the instance is running and has a public IP:

```
ssh -i first_server.pem ec2-user@44.199.252.139
```

Both the AWS CLI and the AWS Management Console (GUI) interact with the same AWS account and AWS resources. Therefore, any EC2 instance launched using the AWS CLI is automatically visible in the EC2 Console of the AWS Management Console.

